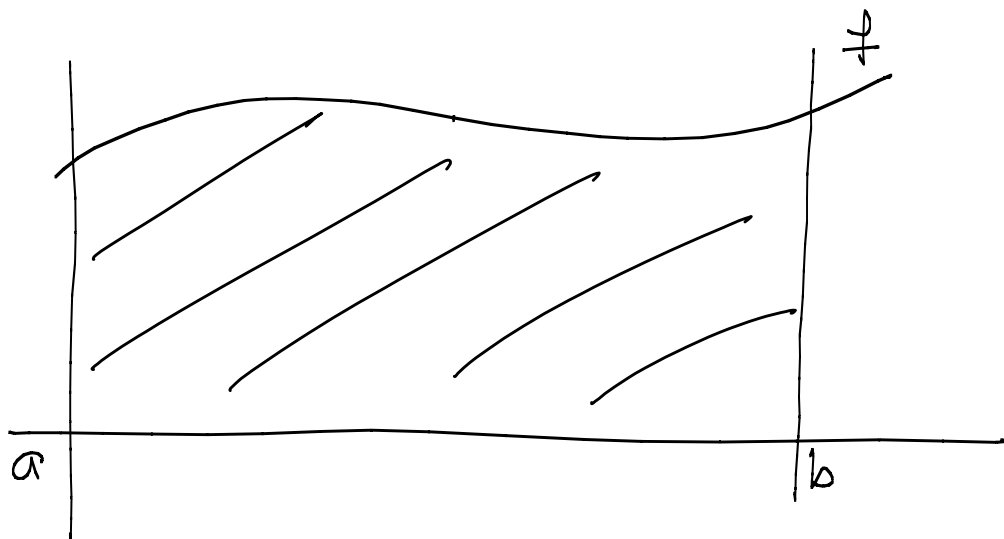


## Unit 5.3 The definite integral

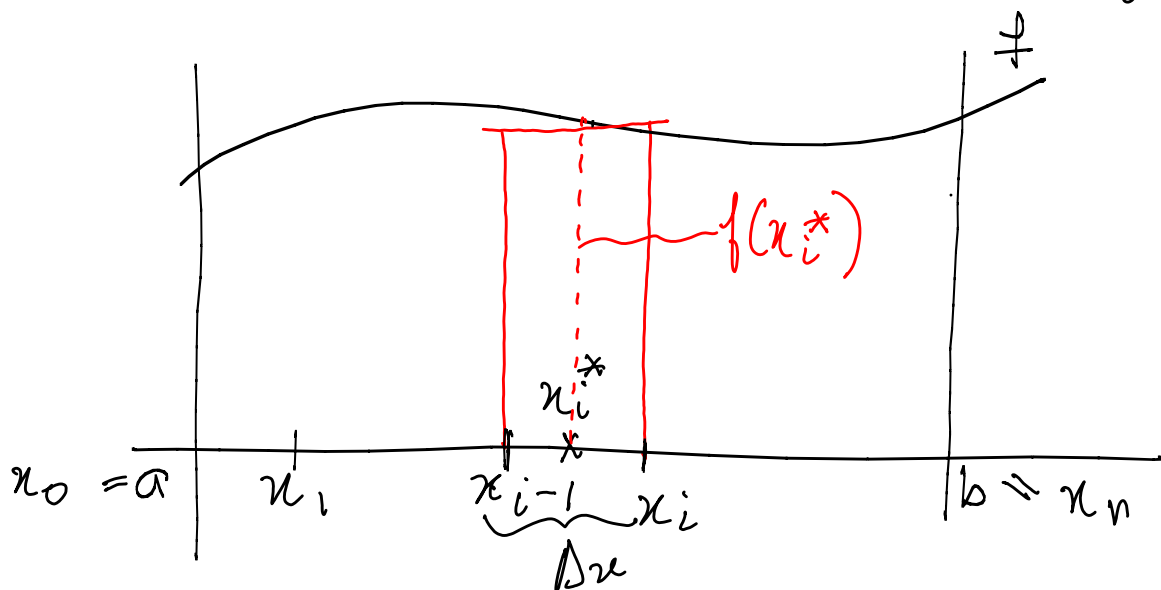
Note Title

2016/05/16

First: The general Riemann sum for the area:



- Say we divide  $[a, b]$  into  $n$  subintervals
- The width of one subinterval is  $\Delta x = \frac{b-a}{n}$
- Consider an interval  $[x_{i-1}, x_i]$
- Pick a random point  $x_i^* \in [x_{i-1}, x_i]$



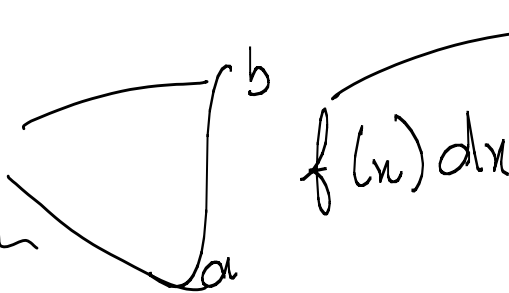
Then the Riemann-sum is

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

And the definite integral is

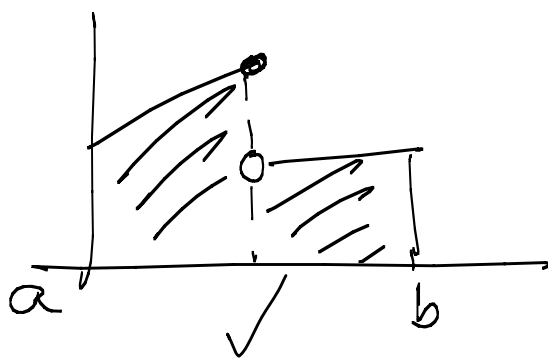
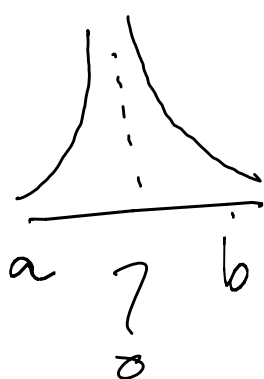
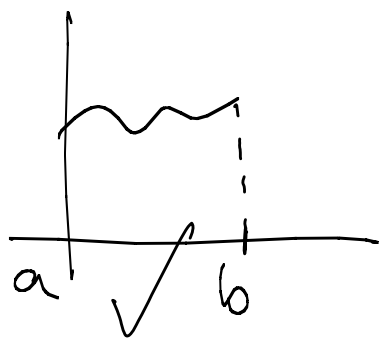
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{f(x_i^*)}_{\text{NB.}} \Delta x$$

If this limit exists we say  $f$  is integrable

limits of integration  integrand

Definite integral.

When do we know that the limit exists?



### Theorem

If  $f$  is continuous on  $[a, b]$  or if  $f$  has a finite number of jump discontinuities then  $f$  is integrable.

## Properties

$$1. \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$2. \int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$3. \int_a^b c f(x) dx = c \int_a^b f(x) dx$$

$$4. \text{ If } f(x) \geq 0 \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq 0$$

$$5. \text{ If } f(x) \leq 0 \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \leq 0$$

$$6. \text{ If } f(x) \geq g(x) \text{ on } [a, b] \text{ then}$$

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

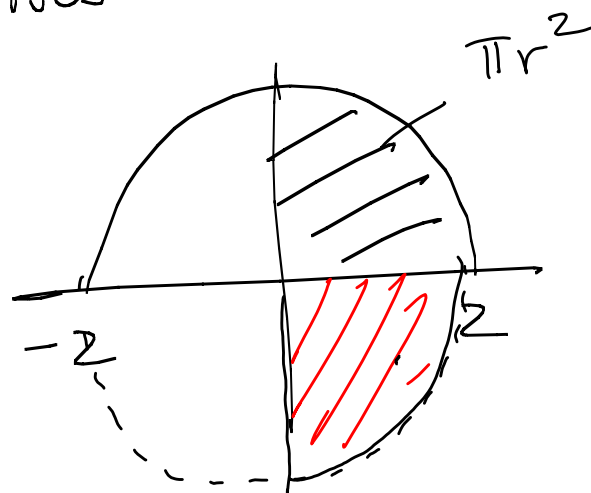
$$7. \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$= \int_a^c f(x) dx$$



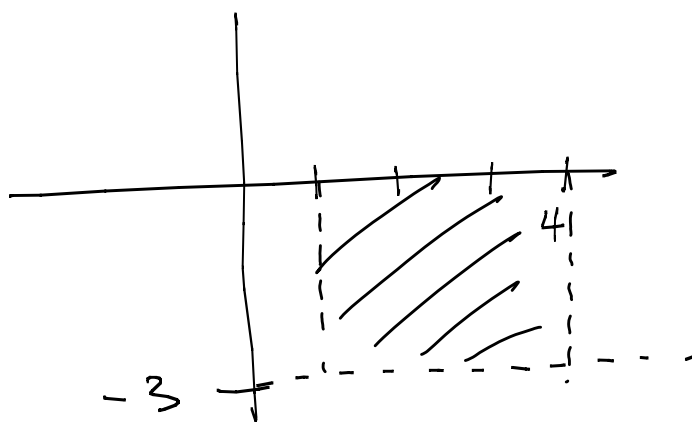
Examples: Calculate without using anti-derivatives

$$1. \int_0^2 \sqrt{4-x^2} dx = \pi$$



$$\int_0^2 -\sqrt{4-x^2} dx = -\pi$$

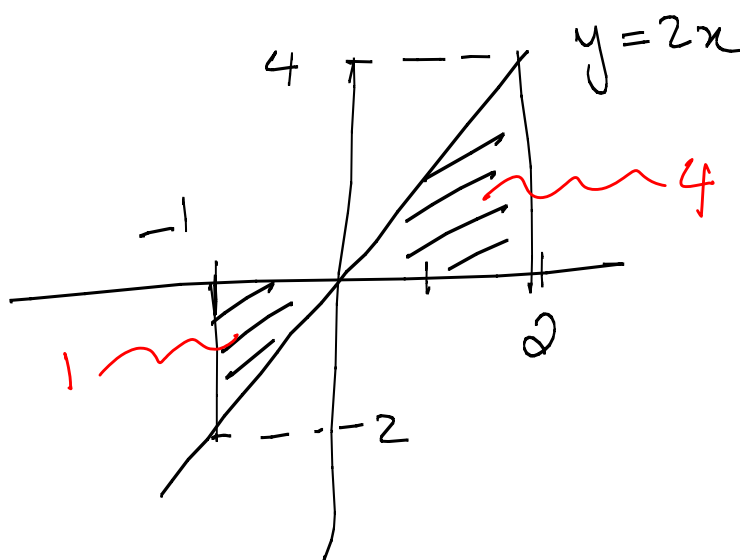
$$2. \int_1^4 -3 dx = -9$$



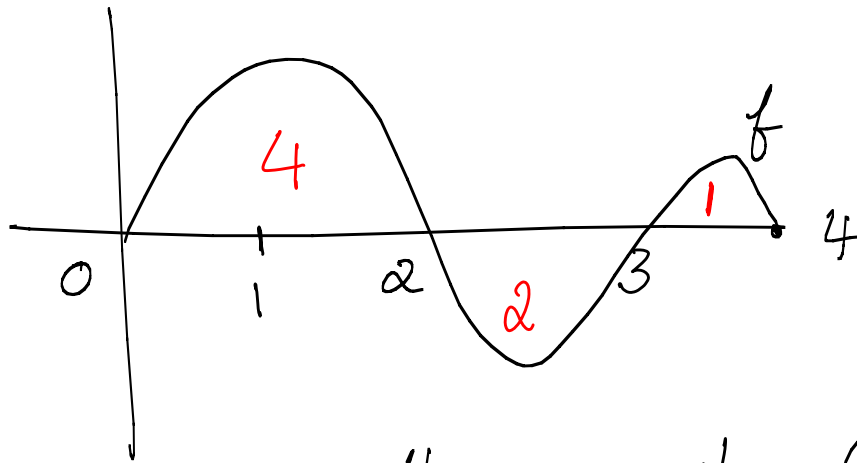
$$3. \int_{-1}^2 2x dx$$

$$= \underbrace{\int_{-1}^0 2x dx}_{-1} + \underbrace{\int_0^2 2x dx}_4$$

$$= 3$$



# Area and integral



$$\bullet \int_0^3 f(x) dx = 2 \quad \bullet \int_2^4 f(x) dx = -1 \quad \bullet \int_0^4 f(x) dx = 3$$

• What is the total area enclosed? 7.

# The Fundamental Theorem of Calculus.

How it works:

$$\begin{aligned}\int_1^2 (2x) dx &= (x^2) \Big|_1^2 \\ &= 4 - (1) \\ &= 3.\end{aligned}$$

$F \rightarrow f \rightarrow f'$

$$\int_a^b f(x) dx = F(b) - F(a).$$

Find:

$$\begin{aligned}\bullet \int_0^4 (x^2 + 4) dx &= \left( \frac{1}{3} x^3 + 4x \right) \Big|_0^4 \\ &= \frac{64}{3} + 16\end{aligned}$$

$$\begin{aligned}\bullet \int_{-1}^2 e^{-x} dx &= -e^{-x} \Big|_{-1}^2 = -e^{-2} + e^1\end{aligned}$$

$$\begin{aligned}\bullet \int_{\pi/2}^{2\pi} \cos x \cdot dx &= \sin x \Big|_{\pi/2}^{2\pi} = 0 - 1 \\ &= -1.\end{aligned}$$